

MERGE SORT

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

void mergesort(int,int);

void merge(int,int,int);

int a[10],n;

void main()

{

    int i,low,high,k;

    printf("MERGE SORT");

    printf("\nEnter the number of elements:");

    scanf("%d",&n);

    printf("\nEnter the elements:\n");

    for(i=1;i<=n;i++)

    {

        scanf("%d",&a[i]);

    }

    low=1;

    high=n;

    mergesort(low,high);

    printf("\nThe sorted list is:");

    for(k=1;k<=n;k++)

    {

        printf("\n%d",a[k]);

    }
```

```

}

void mergesort(int low,int high)
{
    int mid;
    if(low<high)
    {
        mid=((low+high)/2);
        mergesort(low,mid);
        mergesort(mid+1,high);
        merge(low,mid,high);
    }
}

void merge(int low,int mid,int high)
{
    int i,j,h,b[10],k;
    h=low;
    i=low;
    j=mid+1;
    while((h<=mid)&&(j<=high))
    {
        if(a[h]<=a[j])
        {
            b[i]=a[h];
            h=h+1;
        }
        else
        {
            b[i]=a[j];

```

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        j=j+1;
    }
    i=i+1;
}
if(h>mid)
{
    for(k=j;k<=high;k++)
    {
        b[i]=a[k];
        i++;
    }
}
else
{
    for(k=h;k<=mid;k++)
    {
        b[i]=a[k];
        i++;
    }
}
printf("\nsorting process order: ");
for(k=low;k<=high;k++)
{
    a[k]=b[k];
    printf("%d ",a[k]);
}
}
```